Introduction to Scientific Computing

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Motivation

The **three approaches to scientific investigation**

- experimentation (mostly high fidelity, high cost, low accessibility)
- analytic theory (low fidelity, low cost, high accessibility)
- (computer) simulation

Computer simulation promise high accessibility at low cost, but require that a faithful model of the system studied is available.

Motivation

Simulation has always been an integral part of engineering.

Scale model of a hydropower plant

NASA Ames wind tunnel

Computer simulation promise **lower cost** and **faster turnaround time**.

Digital twin as a way to improve and maintain products and equipment.

Pictures from Sebastián Sánchez and NASA Ames Research Center (public domain).

Goal

The goal of this lecture is to give you an idea of what is involved in large scale **numerical simulations** that are used in **science** and **engineering**.

Examples

- Weather forecast
- Aircraft/car design (aerodynamics, elastodynamics, ...)
- Structural integrity of a building (static load, earthquakes, ...)
- Semiconductor manufacturing
- and so on

Goal

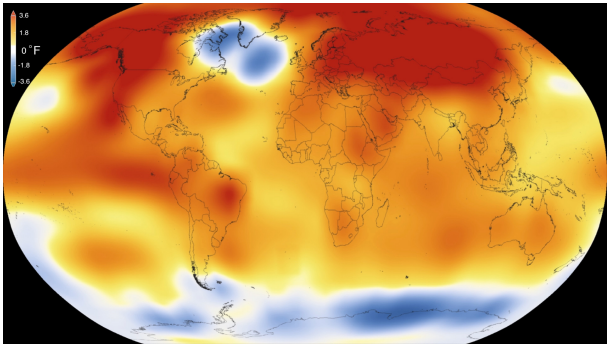
How to go from a physical model to an efficient simulation.

- How a mathematical model (usually a PDE) represents a physical problem.
- How a PDE is represented on a computer (numerical methods).
- Why is a compiled language (C++) used instead of Python or Matlab?
- Familiarize ourselves with programming tools used in high-performance computing.
- Understand the performance characteristics of modern computer systems.
- Understand algorithmic aspects that are essential to perform efficient simulation.
- Visualization and interpretation of the simulation results.

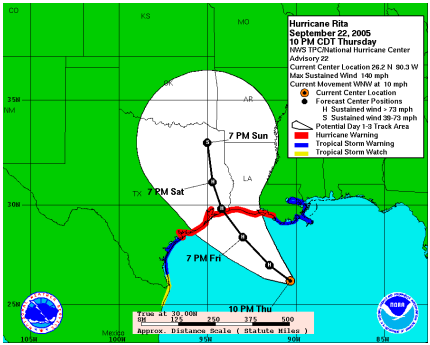
Example 1: Weather forecast

Weather forecast

From weather and climate prediction...



to hurricane forecasting



Pictures from NASA Goddard and NOAA (public domain).

Modeling

The Navier–Stokes equation

$$\begin{aligned}
\partial_t \rho + \nabla \cdot (\rho u) &= 0 \\
\partial_t (\rho u) + \nabla \cdot (\rho u \otimes u) &= -\nabla p + \nabla \cdot \tau + \rho g \\
\partial_t (\rho e) + \nabla \cdot ((\rho e + p) u) &= \nabla \cdot (\tau \cdot u) + \rho (g \cdot u) - \nabla \cdot q
\end{aligned}$$

as a model of the atmosphere (density ρ, velocity u, pressure p, energy e, stress tensor tau, heat flux q, and gravity g).

There are many modeling assumptions

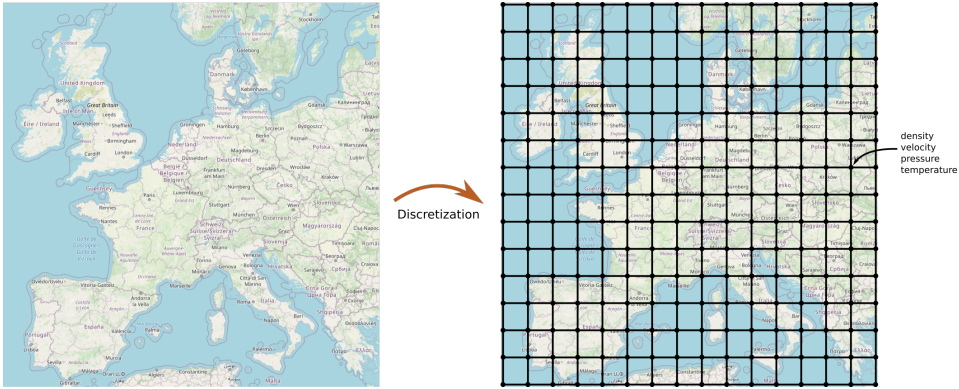
- only one fluid (air)
- no moisture
- ...

Modern weather codes are much more complicated.

Discretization

On a computer there is no continuous space. Everything has to be **discrete**.

Discretization is the process of approximating a continuous problem such that a computer can understand it.



Discretization

There was an effort in the 191x to perform numerical weather prediction by **doing calculations by hand**.

Globe is divided into a grid of size $120x100x5$ and two time steps were were conducted to calculate the weather 6h in advance.

Every *grid point* is assigned a value for density, velocity, pressure, temperature, and water content.

The program failed for a variety of reasons. In Richardson's words

> *Perhaps some day in the dim future it will be possible to advance the computations faster than the weather advances and at a cost less than the saving to mankind due to the information gained. But that is a dream.*

Weather prediction by numerical process, Richardson, 1922.
The Emergence of Numerical Weather Prediction: Richardson's Dream, P. Lynch (`https://maths.ucd.ie/~plynch/Dream/Contents.html`)

Computational cost of (modern) weather prediction

The earth atmosphere is about $5 \cdot 10^8\,\mathrm{km}^2 \times 20\mathrm{km}$.

What is the smallest scale relevant for weather?

- Clouds with a typical size of say 10km across.

Number of grid points in the plane to resolve cloud formation

$$\frac{5 \cdot 10^8}{1 \cdot 1} \times 100 = 5 \cdot 10^{10}.$$

How many time steps do we need for one day

- dictated by how fast the system changes.
- fastest relevant time scale in the system?

Let us assume a maximal wind speed of 100 km/h. Time it takes for air to travel one cell size is 0.6 minutes.

- Would result in ≈ 2500 time steps a day.

Computational cost of weather prediction

To simulate a week in advance (in an hour) we need a computer that can process 10^{15} grid points/hour.

- We commonly take ensemble averages (assume 100 trajectories).

Weather prediction: 10^{17} grid points/hour.
Modern workstation: 10^{13} grid points/hour.

A very optimistic estimate:

- This assumes a perfectly efficient implementation (which is unrealistic)
- This assumes that the explicit Euler method is accurate enough (which is unrealistic)
- This assumes a very simplified model (which is unrealistic)
- This assumes that there is no turbulence (which is unrealistic)

What can be done?

Solution 1: Change the model. Already a simplified model, but we could

- not directly take cloud formation into account (effective model)

Solution 2: Consider better numerical algorithms.

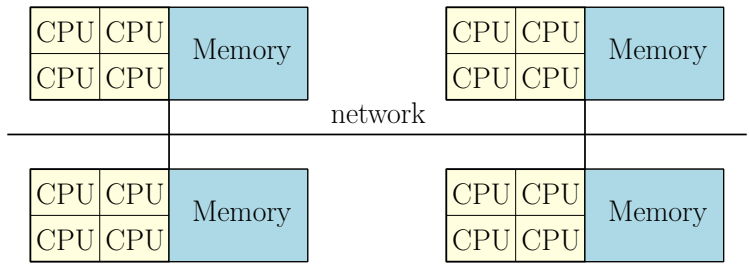
- adaptivity in regions where not much happens
- algorithms that can take time steps that are not restricted by the fastest time scale in the system (implicit methods)

Solution 3: Build bigger computers (supercomputers).

All of the above are essential to modern weather forecasting.

Supercomputers

Modern supercomputers are build by connecting (via a network) a large number of 'commodity computers'.



Top500 list of supercomputers (`http://top500.org`).

Supercomputers

A system in Austria is the Vienna Scientific Cluster 4 (VSC4)

- Performance: $2.7 \cdot 10^{15}$ Flops/s
- Consists of approximately 38k independent processing units.

A larger system is Summit at Oak Ridge National Laboratory (US).

- Performance: 148 Pflops/s
- Consists of approximately 2M independent processing units.

VSC4 (`http://typo3.vsc.ac.at/systems/vsc-4/`)
Summit (`https://www.olcf.ornl.gov/summit/`)
Picture from Carlos Jones, ORNL (CC BY 2.0), `https://tinyurl.com/yxrbf9aw`

Example 2: Aircraft design

Aircraft design

The classic process

- Make a design based on past experience or engineering tables.
- Manufacture a model.
- Test the model in a wind tunnel.
- Based on the obtained measurements refine the design.

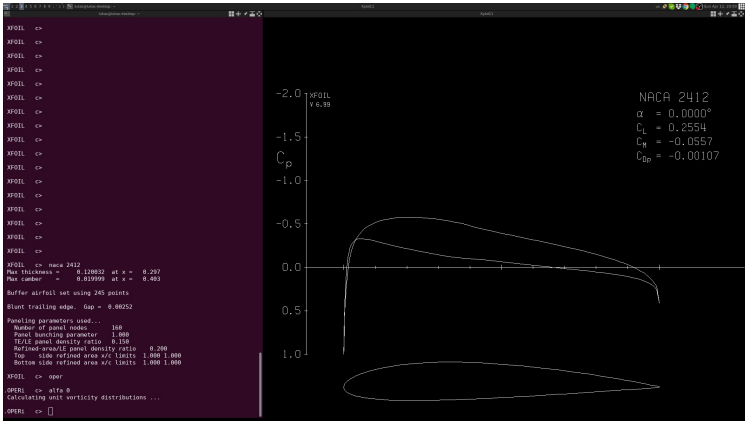
A modern process

- Design is made and evaluated on a computer.
- Often this is done using automatic tools that simplify the design process.
- Make a model to verify the computer simulation.
- Build the aircraft.

Computers have significantly reduced wind tunnel testing.

Panel codes

XFOIL is a program to design subsonic airfoils.

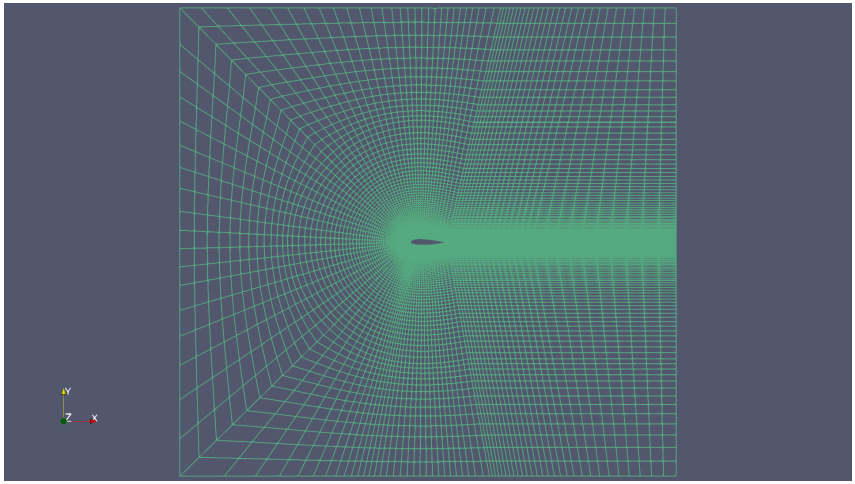


Panel codes are one important tool available to engineers.

- Extremely fast but limited (mostly) to laminar flow.

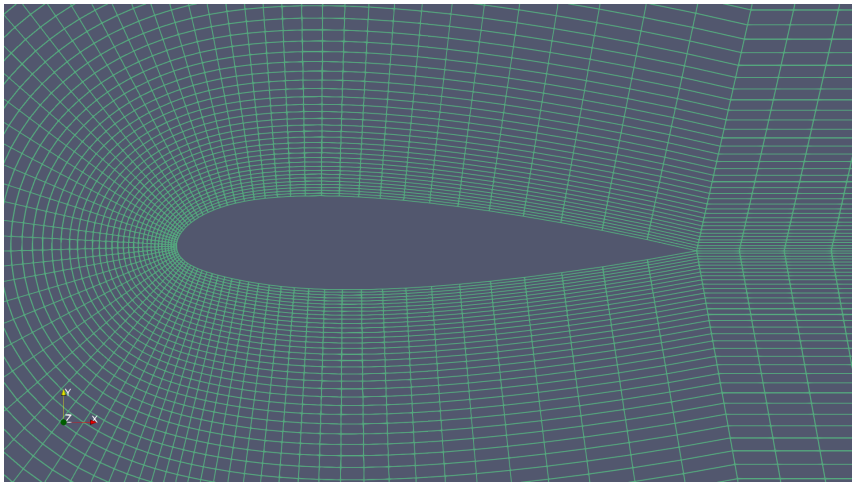
Navier–Stokes solvers

Computational grid for a Navier–Stokes simulation of an airfoil.



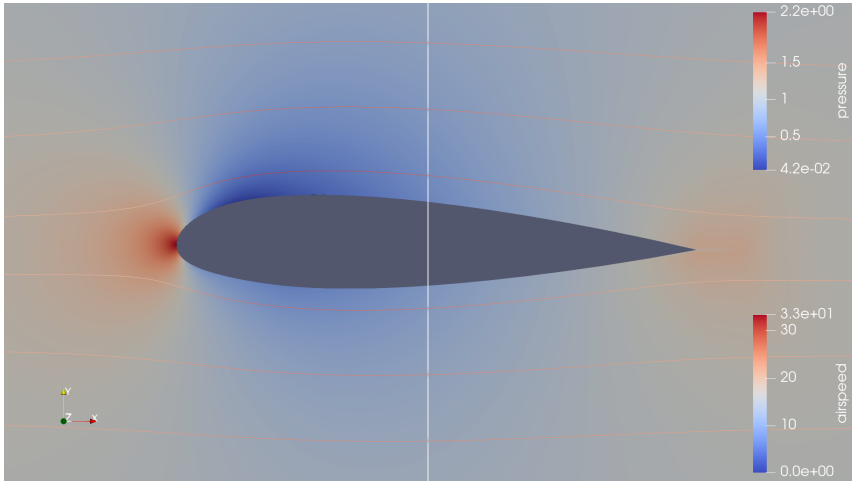
Navier–Stokes solvers

Computational grid for a Navier–Stokes simulation of an airfoil (zoomed in).



Navier–Stokes solvers

OpenFOAM simulation of an airfoil.



Computational fluid dynamics

Computational fluid dynamics spans the range from **panel codes (very fast)** to **full three-dimensional Navier–Stokes equations (extremely expensive)**.

- Tradeoff between fast turnaround time and high-fidelity results.

There is ample evidence that the Navier–Stokes equation faithfully resolves fluid flow, but performing a full Navier–Stokes simulation can be extremely expensive.

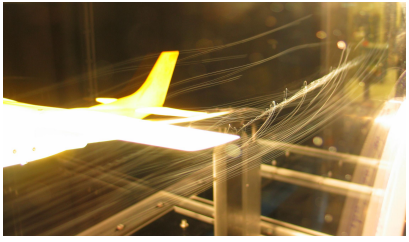
A range of simplified models are available that offer faster time to solution.

- Two-dimensional models.
- Inviscid solvers (e.g. panel codes).
- Incompressible models (no propagation of sound waves).
- Turbulence models.

An economic argument

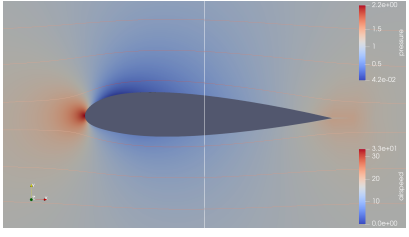
Wind tunnel:

- Building models is expensive
- Using wind tunnels is expensive
- Both take a significant amounts of time
- Limitations on what aspects of the flow field can be measured (simultaneously)



Computer simulation:

- Turnaround time can be extremely fast
- Allow for the use of automated design tools
- Provide the entire flow field at no extra cost
- No safety concerns.



Picture from BenFrantzDale (CC BY-SA 3.0), https://tinyurl.com/y2k2okkg

Scientific computing as a discipline

Scientific computing

Scientific computing is concerned with **selecting appropriate models and numerical algorithms** to solve problems from science and engineering and **their efficient implementation** on modern computer systems.

High-performance computing is concerned with systems that offer much more computational power than a single desktop or workstation.

Assumes that we have good models of the systems we investigate.

Requires expertise distinct from domain science.

- Interplay between mathematical models and numerical algorithms.
- Optimization of computer programs on modern hardware architectures.
- Parallelization of computer codes.
- Working in close collaboration with domain scientists.

A broader economic argument

Computer simulations can still be expensive.

- The cost is usually estimated at approximately 0.10 EUR per core-hour.

Running a simulation with 10k cores for three days: 72k EUR.

- For a high fidelity simulation this is not uncommon.

Summit had a procurement cost of 200M $ and draws 13 MW of power.

Optimizing a program can have a huge impact.

- Even if the performance in only improved by a small amount.

In scientific computing, performance matters!